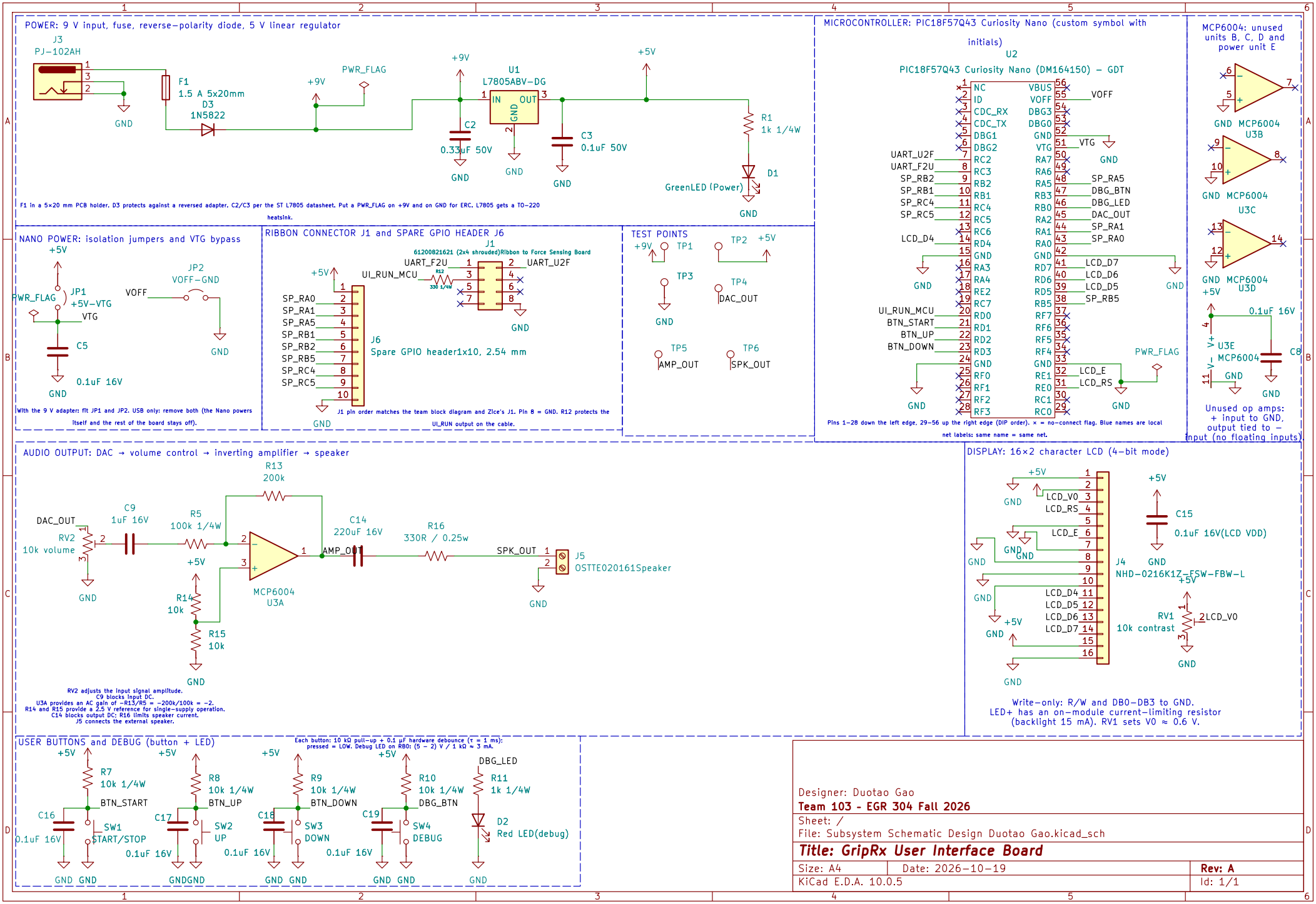

POWER: 9 V input, fuse, reverse-polarity diode, 5 V linear regulator
J3
PJ-102AH
F1
1.5 A 5x20mm
D3
1N5822
PWR_FLAG
U1
L7805ABV-DG
IN
OUT
GND
C2
0.33uF 50V
C3
0.1uF 50V
R1
1k 1/4W
D1
GreenLED (Power)
+9V
+5V
GND
F1 in a 5×20 mm PCB holder. D3 protects against a reversed adapter. C2/C3 per the ST L7805 datasheet. Put a PWR_FLAG on +9V and on GND for ERC. L7805 gets a TO-220 heatsink.
MICROCONTROLLER: PIC18F57Q43 Curiosity Nano (custom symbol with initials)
U2
PIC18F57Q43 Curiosity Nano (DM164150) – GDT
NC
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBG0
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
RE1
RE0
RC1
RC0
UART_U2F
UART_F2U
SP_RB2
SP_RB1
SP_RC4
SP_RC5
LCD_D4
UI_RUN_MCU
BTN_START
BTN_UP
BTN_DOWN
SP_RA5
DBG_BTN
DBG_LED
DAC_OUT
SP_RA1
SP_RA0
LCD_D7
LCD_D6
LCD_D5
SP_RB5
LCD_E
LCD_RS
PWR_FLAG
Pins 1–28 down the left edge, 29–56 up the right edge (DIP order). × = no-connect flag. Blue names are local net labels; same name = same net.
MCP6004: unused units B, C, D and power unit E
GND MCP6004
U3B
U3C
U3D
U3E
+5V
0.1uF 16V
C8
Unused op amps: + input to GND, output tied to – input (no floating inputs).
NANO POWER: isolation jumpers and VTG bypass
JP1
+5V–VTG
JP2
VOFF–GND
VOFF
VTG
C5
0.1uF 16V
With the 9 V adapter: fit JP1 and JP2. USB only: remove both (the Nano powers itself and the rest of the board stays off).
RIBBON CONNECTOR J1 and SPARE GPIO HEADER J6
J1
61200821621 (2x4 shrouded)Ribbon to Force Sensing Board
UART_F2U
UART_U2F
UI_RUN_MCU
R12
330 1/4W
J6
Spare GPIO header1x10, 2.54 mm
SP_RA0
SP_RA1
SP_RA5
SP_RB1
SP_RB2
SP_RB5
SP_RC4
SP_RC5
J1 pin order matches the team block diagram and Zlce's J1. Pin 8 = GND. R12 protects the UI_RUN output on the cable.
TEST POINTS
TP1
TP2
TP3
TP4
TP5
TP6
DAC_OUT
AMP_OUT
SPK_OUT
AUDIO OUTPUT: DAC → volume control → inverting amplifier → speaker
DAC_OUT
C9
1uF 16V
RV2
10k volume
R5
100k 1/4W
R13
200k
C14
220uF 16V
R16
330R / 0.25w
SPK_OUT
J5
OSTTE020161Speaker
MCP6004
U3A
R14
10k
R15
10k
RV2 adjusts the input signal amplitude.
C9 blocks input DC.
U3A provides an AC gain of −R13/R5 = −200k/100k = −2.
R14 and R15 provide a 2.5 V reference for single-supply operation.
C14 blocks output DC; R16 limits speaker current.
J5 connects the external speaker.
DISPLAY: 16×2 character LCD (4-bit mode)
J4
NHD-0216K1Z-FSW-FBW-L
LCD_VO
LCD_RS
LCD_E
LCD_D4
LCD_D5
LCD_D6
LCD_D7
C15
0.1uF 16V(LCD VDD)
RV1
10k contrast
Write-only: R/W and DB0–DB3 to GND.
LED+ has an on-module current-limiting resistor (backlight 15 mA). RV1 sets V0 ≈ 0.6 V.
USER BUTTONS and DEBUG (button + LED)
Each button: 10 kΩ pull-up + 0.1 µF hardware debounce (τ = 1 ms): pressed = LOW. Debug LED on RB0: (5 − 2) V / 1 kΩ ≈ 3 mA.
R7
10k 1/4W
BTN_START
C16
0.1uF 16V
SW1
START/STOP
R8
10k 1/4W
BTN_UP
C17
SW2
UP
R9
10k 1/4W
BTN_DOWN
C18
SW3
DOWN
R10
10k 1/4W
DBG_BTN
C19
SW4
DEBUG
DBG_LED
R11
1k 1/4W
D2
Red LED(debug)
Designer: Duotao Gao
Team 103 - EGR 304 Fall 2026
Sheet: /
File: Subsystem Schematic Design Duotao Gao.kicad_sch
Title: GripRx User Interface Board
Size: A4
Date: 2026-10-19
Rev: A
KiCad E.D.A. 10.0.5
Id: 1/1